```
!pip install transformers datasets torch scikit-learn
```

```
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dis
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.1
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist
Requirement already satisfied: pyarrow>=15.0.0 in /usr/local/lib/python3.12/
Requirement already satisfied: dill<0.3.9,>=0.3.0 in /usr/local/lib/python3
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-pack
Requirement already satisfied: requests>=2.32.2 in /usr/local/lib/python3.12
Requirement already satisfied: xxhash in /usr/local/lib/python3.12/dist-pack
Requirement already satisfied: multiprocess<0.70.17 in /usr/local/lib/python
Requirement already satisfied: fsspec<=2025.3.0,>=2023.1.0 in /usr/local/lib
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/p
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/di
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-pack
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dis
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/di
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python
Requirement already satisfied: aiohttp!=4.0.0a0,!=4.0.0a1 in /usr/local/lib/
Requirement already satisfied: hf-xet<2.0.0,>=1.2.0 in /usr/local/lib/python
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packa
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/py
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dis
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/pyth
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dis
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/d
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/pyt
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.12
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/di
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.1
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.1
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packa
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/di
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dis
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3
```

```
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist
```

```
import torch
from datasets import load_dataset
from transformers import AutoTokenizer, AutoModelForSequenceClassification
from transformers import TrainingArguments, Trainer
from sklearn.metrics import accuracy_score, precision_recall_fscore_support
```

```
dataset = load_dataset("imdb")

train_dataset = dataset['train']
test_dataset = dataset['test']
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: Us
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings ta
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access p
  warnings.warn(
README.md:       7.81k/? [00:00<00:00, 463kB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated
plain_text/train-00000-of-00001.parquet: 100%  21.0M/21.0M [00:00<00:00, 22.0MB/s]
plain_text/test-00000-of-00001.parquet: 100%  20.5M/20.5M [00:00<00:00, 131MB/s]
plain_text/unsupervised-00000-of-00001.p(…): 100%  42.0M/42.0M [00:00<00:00, 163MB/s]
Generating train split: 100%  25000/25000 [00:00<00:00, 54661.50 examples/s]
Generating test split: 100%  25000/25000 [00:00<00:00, 64417.24 examples/s]
Generating unsupervised split: 100%  50000/50000 [00:00<00:00, 75163.81 examples/s]
```

```
print(train_dataset[0])
```

```
{'text': 'I rented I AM CURIOUS-YELLOW from my video store because of all the
```

```
tokenizer = AutoTokenizer.from_pretrained("distilbert-base-uncased")
```

```
config.json: 100%                                    483/483 [00:00<00:00, 14.7kB/s]

tokenizer_config.json: 100%                          48.0/48.0 [00:00<00:00, 627B/s]

vocab.txt:        232k/? [00:00<00:00, 1.23MB/s]

tokenizer.json:       466k/? [00:00<00:00, 3.88MB/s]
```

```
def tokenize_function(examples):
    return tokenizer(examples["text"], padding="max_length", truncation=True

train_dataset = train_dataset.map(tokenize_function, batched=True)
test_dataset = test_dataset.map(tokenize_function, batched=True)
```

```
Map: 100%                                 25000/25000 [01:15<00:00, 226.42 examples/s]

Map: 100%                                 25000/25000 [01:06<00:00, 374.96 examples/s]
```

```
train_dataset.set_format(type="torch", columns=["input_ids", "attention_mask
test_dataset.set_format(type="torch", columns=["input_ids", "attention_mask"
```

```
model = AutoModelForSequenceClassification.from_pretrained(
    "distilbert-base-uncased",
    num_labels=2
)
```

```
model.safetensors: 100%                              268M/268M [00:03<00:00, 148MB/s]

Loading weights: 100%  100/100 [00:00<00:00, 338.41it/s, Materializing param=distilbert.transforr

DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased
Key                      | Status     |
-------------------------+------------+-
vocab_layer_norm.weight  | UNEXPECTED |
vocab_layer_norm.bias    | UNEXPECTED |
vocab_transform.bias     | UNEXPECTED |
vocab_projector.bias     | UNEXPECTED |
vocab_transform.weight   | UNEXPECTED |
pre_classifier.weight    | MISSING    |
classifier.bias          | MISSING    |
classifier.weight        | MISSING    |
pre_classifier.bias      | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture
- MISSING   :those params were newly initialized because missing from the che
```

```
def compute_metrics(pred):
    labels = pred.label_ids
    preds = pred.predictions.argmax(-1)

    precision, recall, f1, _ = precision_recall_fscore_support(labels, preds
```

```
    acc = accuracy_score(labels, preds)

    return {
        'accuracy': acc,
        'f1': f1,
        'precision': precision,
        'recall': recall
    }
```

```
from transformers.training_args import IntervalStrategy

training_args = TrainingArguments(
    output_dir="./results",
    # Temporarily removed evaluation_strategy due to persistent TypeError
    # evaluation_strategy=IntervalStrategy.EPOCH,
    learning_rate=2e-5,
    per_device_train_batch_size=8,
    per_device_eval_batch_size=8,
    num_train_epochs=2,
    weight_decay=0.01
)
```

```
trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=train_dataset,
    eval_dataset=test_dataset,
    compute_metrics=compute_metrics
)

trainer.train()
```

```
/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:775: U
  super().__init__(loader)
```

[160/6250 1:11:46 < 46:06:33, 0.04 it/s, Epoch 0.05/2]

Step	Training Loss

```
results = trainer.evaluate()
print(results)
```

```
text = "The movie was absolutely amazing!"

inputs = tokenizer(text, return_tensors="pt", truncation=True, padding=True)
outputs = model(**inputs)
```

```
prediction = torch.argmax(outputs.logits)

if prediction == 1:
    print("Positive Sentiment")
else:
    print("Negative Sentiment")
```

```
Positive Sentiment
```